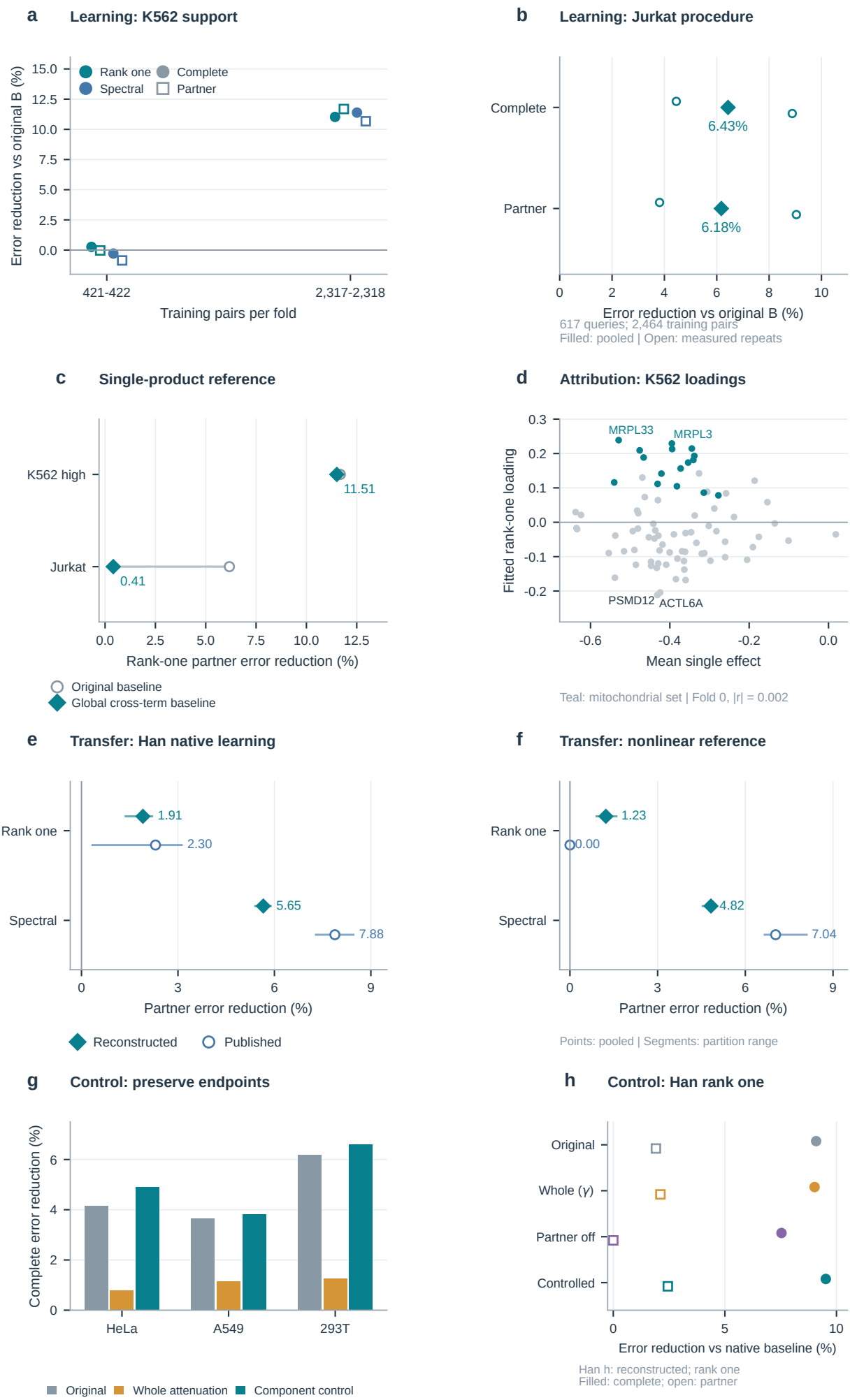
a
Learning: K562 support
Rank one
Complete
Spectral
Partner
Error reduction vs original B (%)
15.0
12.5
10.0
7.5
5.0
2.5
0.0
421-422
2,317-2,318
Training pairs per fold
b
Learning: Jurkat procedure
Complete
Partner
6.43%
6.18%
0
2
4
6
8
10
Error reduction vs original B (%)
617 queries; 2,464 training pairs
Filled: pooled | Open: measured repeats
c
Single-product reference
K562 high
Jurkat
11.51
0.41
0.0
2.5
5.0
7.5
10.0
12.5
Rank-one partner error reduction (%)
Original baseline
Global cross-term baseline
d
Attribution: K562 loadings
MRPL33
MRPL3
PSMD12
ACTL6A
Fitted rank-one loading
0.3
0.2
0.1
0.0
-0.1
-0.2
-0.6
-0.4
-0.2
0.0
Mean single effect
Teal: mitochondrial set | Fold 0, |r| = 0.002
e
Transfer: Han native learning
Rank one
Spectral
1.91
2.30
5.65
7.88
0
3
6
9
Partner error reduction (%)
Reconstructed
Published
f
Transfer: nonlinear reference
Rank one
Spectral
1.23
0.00
4.82
7.04
0
3
6
9
Partner error reduction (%)
Points: pooled | Segments: partition range
g
Control: preserve endpoints
Complete error reduction (%)
6
4
2
0
HeLa
A549
293T
Original
Whole attenuation
Component control
h
Control: Han rank one
Original
Whole (γ)
Partner off
Controlled
0
5
10
Error reduction vs native baseline (%)
Han h: reconstructed; rank one
Filled: complete; open: partner